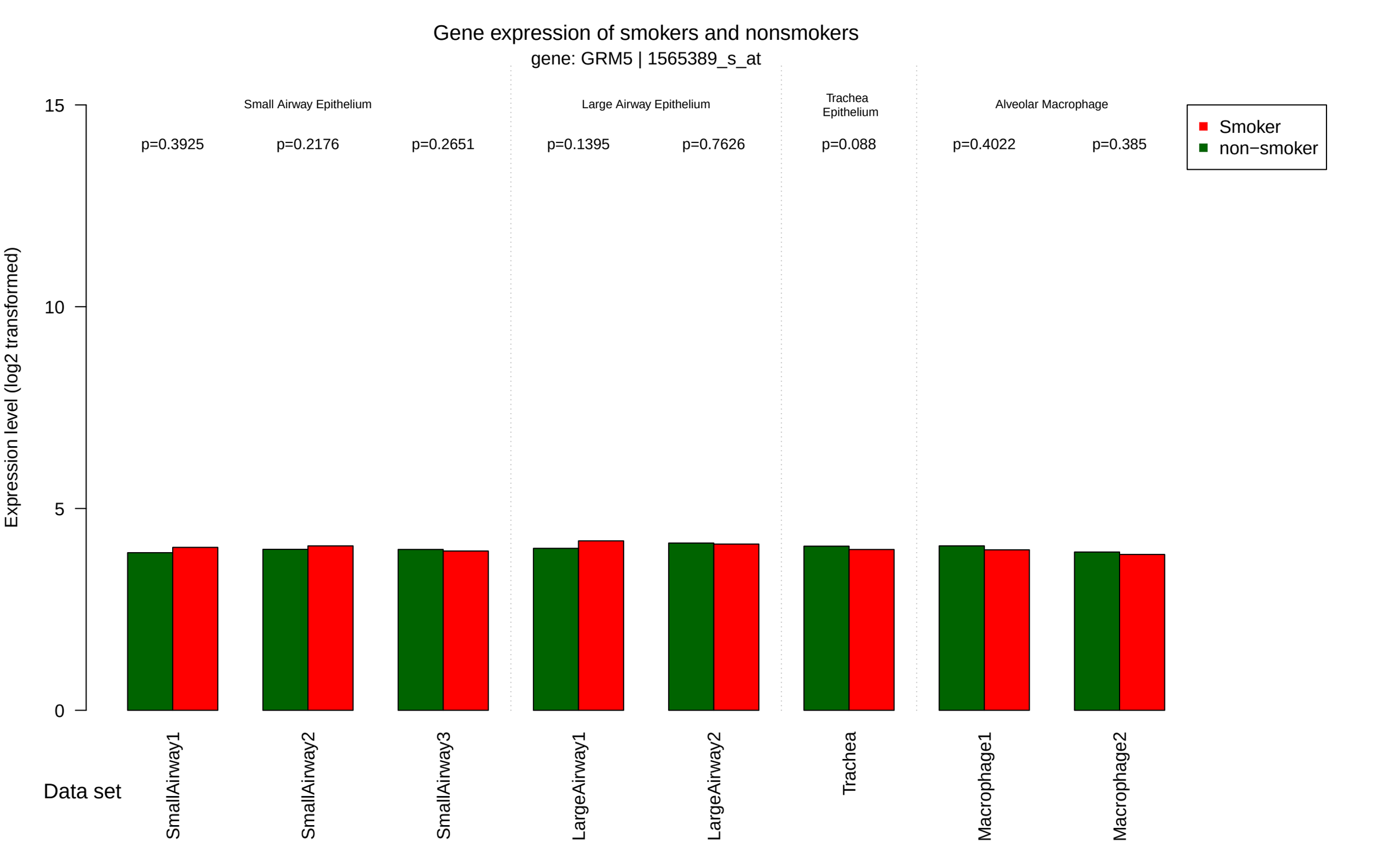
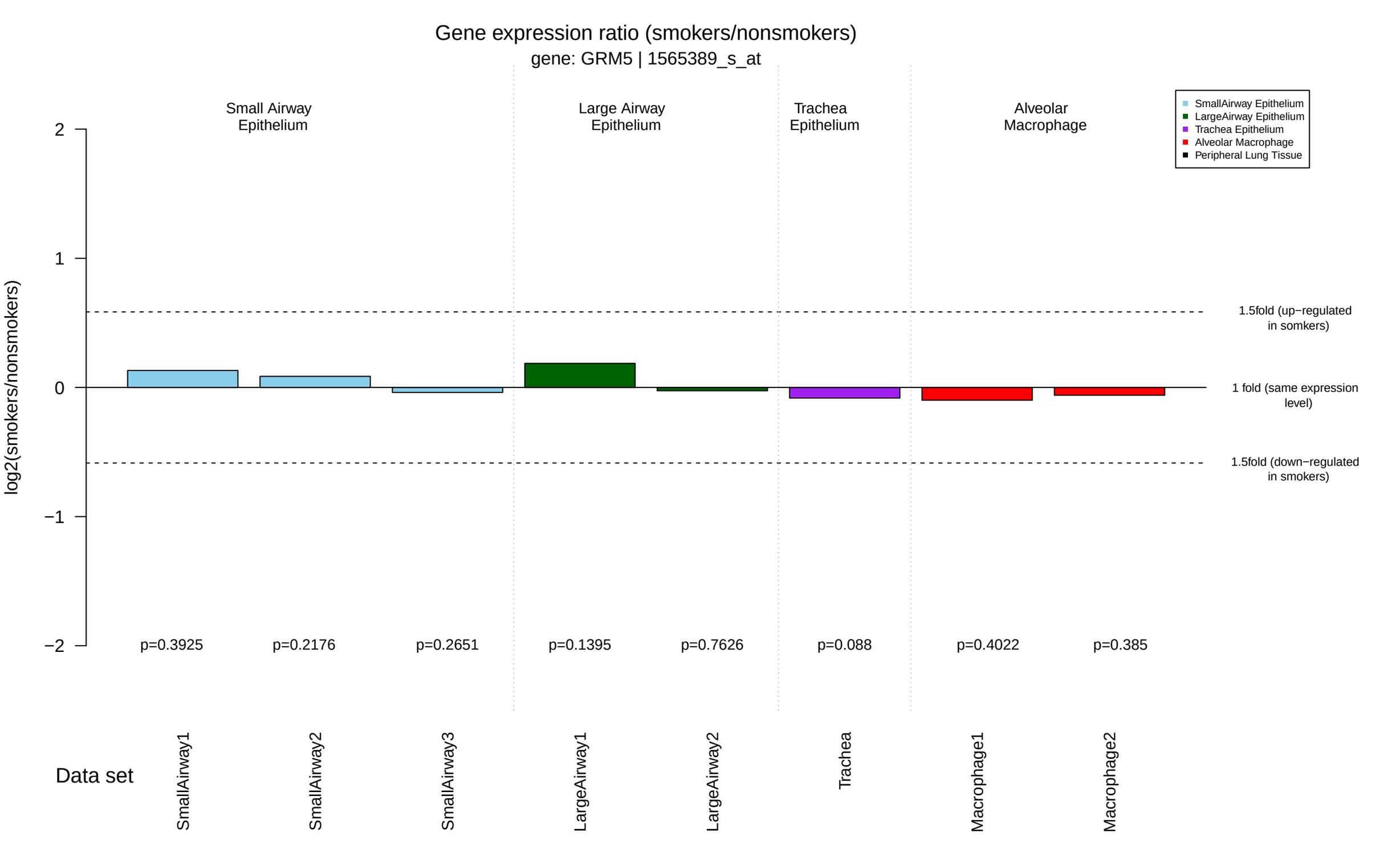
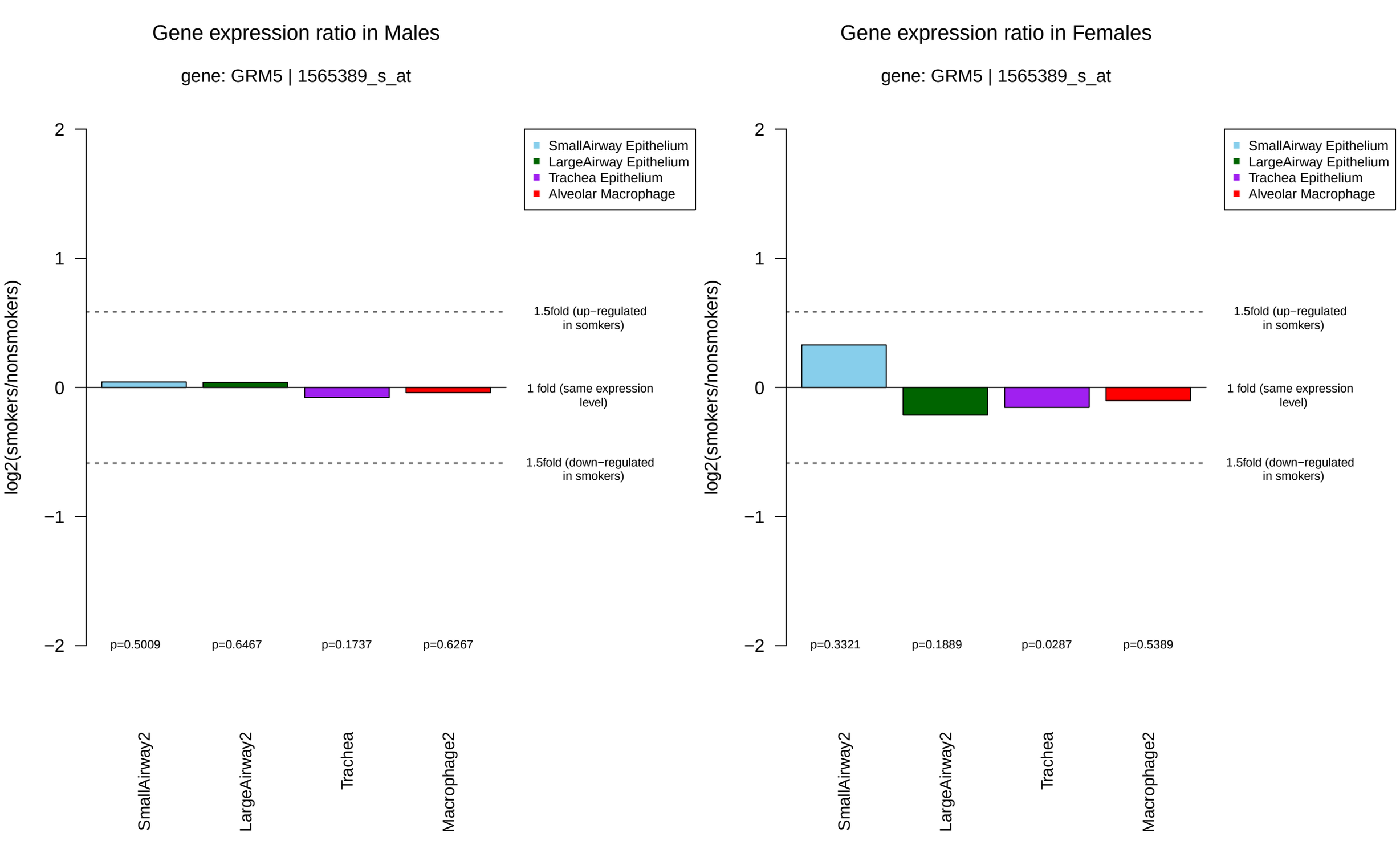
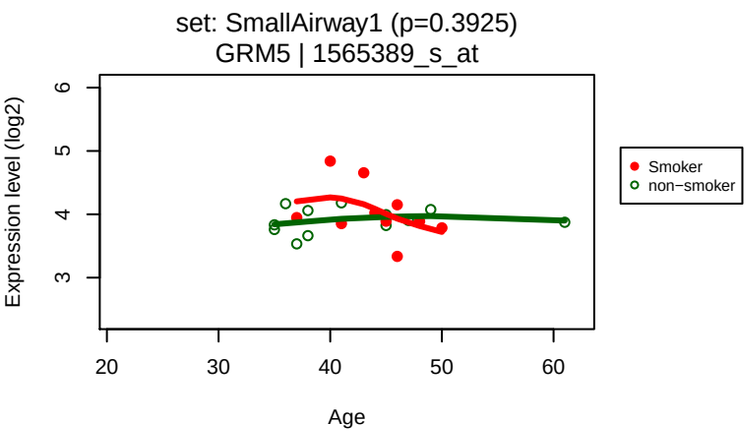
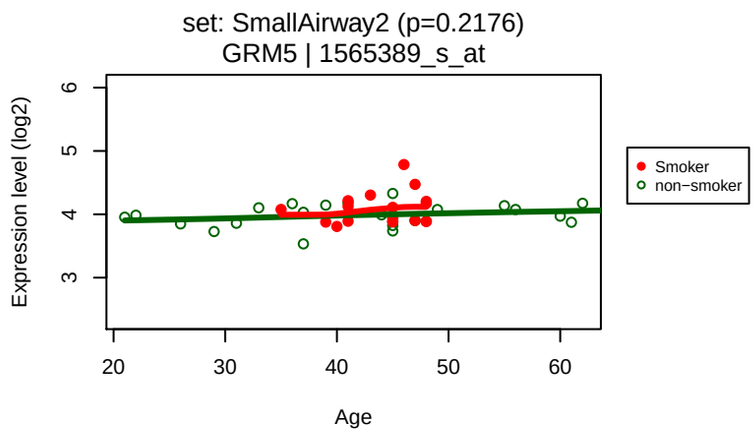
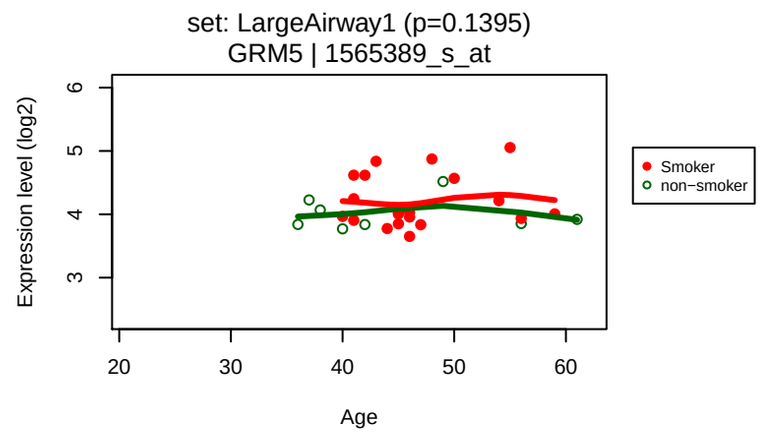

Gene expression of smokers and nonsmokers
gene: GRM5 | 1565389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3925
p=0.2176
p=0.2651
p=0.1395
p=0.7626
p=0.088
p=0.4022
p=0.385
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

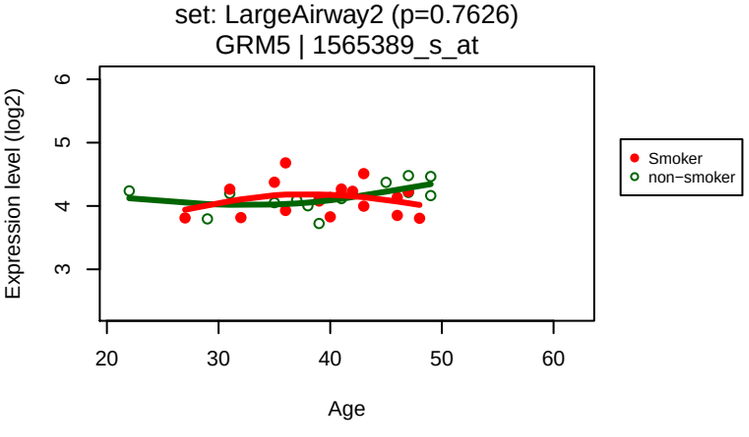

Gene expression ratio (smokers/nonsmokers)
gene: GRM5 | 1565389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3925
p=0.2176
p=0.2651
p=0.1395
p=0.7626
p=0.088
p=0.4022
p=0.385
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

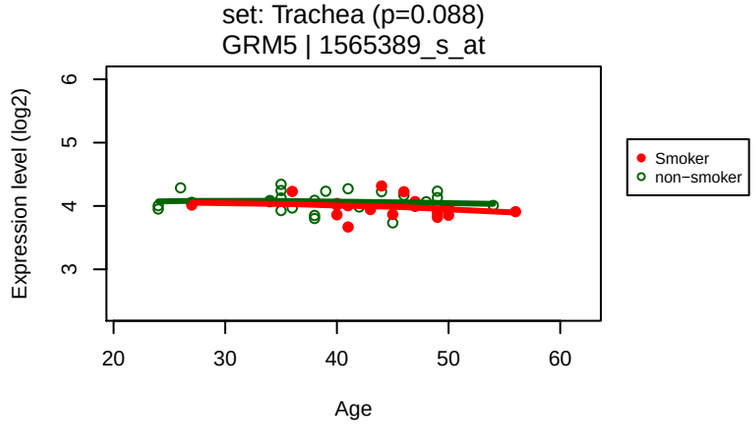

Gene expression ratio in Males
gene: GRM5 | 1565389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5009
p=0.6467
p=0.1737
p=0.6267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM5 | 1565389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3321
p=0.1889
p=0.0287
p=0.5389
SmallAirway2
LargeAirway2
Trachea
Macrophage2

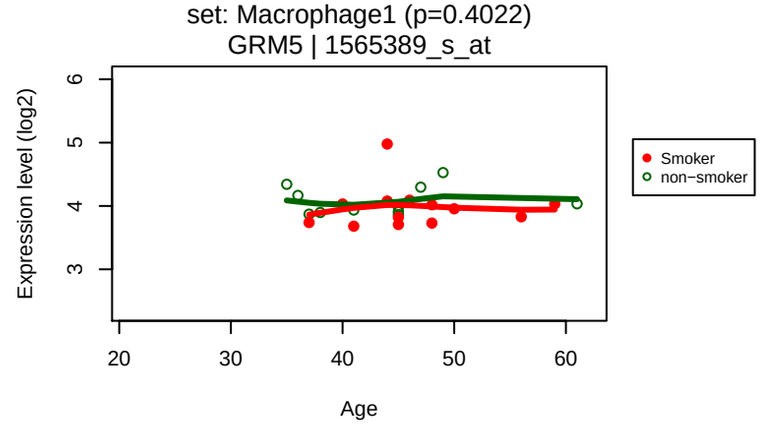

set: SmallAirway1 (p=0.3925)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker

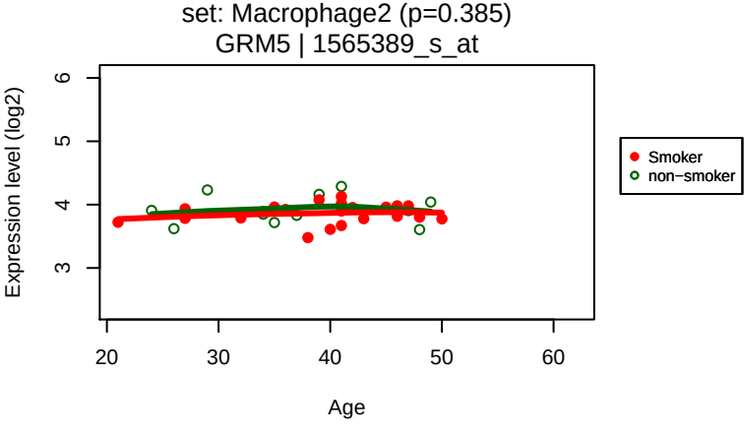

set: SmallAirway2 (p=0.2176)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker

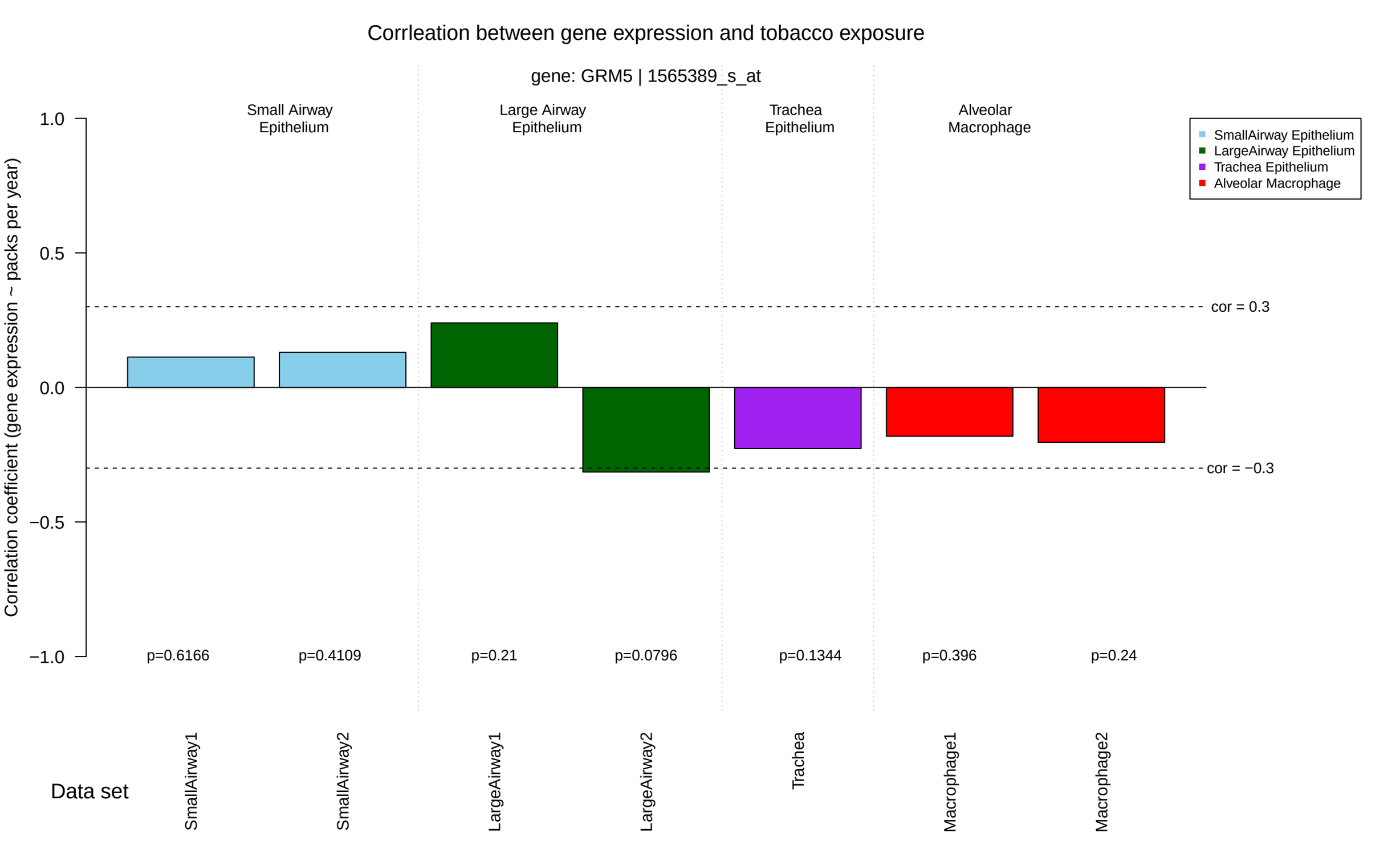

set: LargeAirway1 (p=0.1395)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7626)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.088)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4022)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.385)
GRM5 | 1565389_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM5 | 1565389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6166
p=0.4109
p=0.21
p=0.0796
p=0.1344
p=0.396
p=0.24
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2